KAIZEN MATHS

Lesson starter

Simultaneous Equations

Substitution Method

Name _____

S1

Solve the system:

$y = -4x - 6$

$3x + 2y = -27$

S2

Solve the system:

$3x - 4y = 2$

$4x - 2y = 6$

S3

Solve the system:

$y = 5x + 4$

$6x + 4y = 94$

S4

Solve the system:

$4x - 4y = -4$

$3x - 2y = 0$

Show your working

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ALG-BE09-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Simultaneous Equations

Substitution Method

Name _____

E1

Solve the system:

$y = 5x + 6$

$2x + 2y = 36$

E2

Solve the system:

$2x + 5y = 13$

$3x + 5y = 17$

E3

Solve the system:

$y = -5x - 3$

$6x + 4y = -54$

ALG-BE09-L1 · E

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Exit ticket

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ALG-BE09-L1 · E

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ALG-BE09-L1 · E

Show your working

Teacher answers & working

Simultaneous Equations · Substitution Method

ALG-BE09-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Solve the system:

$y = -4x - 6$

$3x + 2y = -27$

Answer:

x = 3,\; y = -18

Working:

Step 1: Write the system.

$$\begin{cases} y = -4x - 6 \\ 3x + 2y = -27 \end{cases}$$

Step 2: Substitute the expression for y into the second equation.

3x + 2\left(-4x - 6\right) = -27

Step 3: Expand and collect like terms.

3x + -8x - 12 = -27

-5x - 12 = -27

Step 4: Move the constant term and solve for x.

-5x = -27 + 12 = -15

x = 3 = 3

Step 5: Substitute x back into y = mx + b.

$y = -4 \times 3 - 6 = -18$

Step 6: Check in the second equation.

$3 \times 3 + 2 \times -18 = -27$ ✓

Teacher answers & working

Simultaneous Equations · Substitution Method

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S2

Solve the system:

$3x - 4y = 2$

$4x - 2y = 6$

Answer:

$x = 2, \; y = 1$

Working:

Step 1: Write the system.

$$\begin{cases} 3x - 4y = 2 \\ 4x - 2y = 6 \end{cases}$$

Step 2: Rearrange equation (1) to make y the subject.

$-4y = 2 - 3x$

$y = \frac{3}{4}x + \frac{-1}{2}$

Step 3: Substitute into equation (2).

$4x + -2\left(\frac{3}{4}x + \frac{-1}{2}\right) = 6$

Step 4: Expand, collect, and solve.

$4x + \frac{-3}{2}x + 1 = 6$

$x = 2 = 2$

Step 5: Substitute x back for y.

$y = \frac{3}{4} \times 2 + \frac{-1}{2} = 1 = 1$

Step 6: Check both original equations.

$3 \times 2 + -4 \times 1 = 2$ ✓

$4 \times 2 + -2 \times 1 = 6$ ✓

Teacher answers & working

Simultaneous Equations · Substitution Method

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S3

Solve the system:

$y = 5x + 4$

$6x + 4y = 94$

Answer:

$x = 3, \; y = 19$

Working:

Step 1: Write the system.

$$\begin{cases} y = 5x + 4 \\ 6x + 4y = 94 \end{cases}$$

Step 2: Substitute the expression for y into the second equation.

6x + 4\left(5x + 4\right) = 94

Step 3: Expand and collect like terms.

$6x + 20x + 16 = 94$

$26x + 16 = 94$

Step 4: Move the constant term and solve for x.

$26x = 94 - 16 = 78$

$x = 3 = 3$

Step 5: Substitute x back into y = mx + b.

$y = 5 \times 3 + 4 = 19$

Step 6: Check in the second equation.

$6 \times 3 + 4 \times 19 = 94$ ✓

Teacher answers & working

Simultaneous Equations · Substitution Method

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S4

Solve the system:

$4x - 4y = -4$

$3x - 2y = 0$

Answer:

$x = 2, \; y = 3$

Working:

Step 1: Write the system.

$$\begin{cases} 4x - 4y = -4 \\ 3x - 2y = 0 \end{cases}$$

Step 2: Rearrange equation (1) to make y the subject.

-4y = -4 - 4x

$y = x + 1$

Step 3: Substitute into equation (2).

3x + -2\left(x + 1\right) = 0

Step 4: Expand, collect, and solve.

$3x + -2x + -2 = 0$

$x = 2 = 2$

Step 5: Substitute x back for y.

$y = 1 \times 2 + 1 = 3 = 3$

Step 6: Check both original equations.

$4 \times 2 + -4 \times 3 = -4$ ✓

$3 \times 2 + -2 \times 3 = 0$ ✓

Teacher answers & working

Simultaneous Equations · Substitution Method

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E1

Solve the system:

$y = 5x + 6$

$2x + 2y = 36$

Answer:

$x = 2, \; y = 16$

Working:

Step 1: Write the system.

$$\begin{cases} y = 5x + 6 \\ 2x + 2y = 36 \end{cases}$$

Step 2: Substitute the expression for y into the second equation.

2x + 2\left(5x + 6\right) = 36

Step 3: Expand and collect like terms.

$2x + 10x + 12 = 36$

$12x + 12 = 36$

Step 4: Move the constant term and solve for x.

$12x = 36 - 12 = 24$

$x = 2 = 2$

Step 5: Substitute x back into y = mx + b.

$y = 5 \times 2 + 6 = 16$

Step 6: Check in the second equation.

$2 \times 2 + 2 \times 16 = 36$ ✓

Teacher answers & working

Simultaneous Equations · Substitution Method

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E2

Solve the system:

$2x + 5y = 13$

$3x + 5y = 17$

Answer:

$x = 4, \; y = 1$

Working:

Step 1: Write the system.

$$\begin{cases} 2x + 5y = 13 \\ 3x + 5y = 17 \end{cases}$$

Step 2: Rearrange equation (1) to make y the subject.

$5y = 13 - 2x$

$y = \frac{-2}{5}x + \frac{13}{5}$

Step 3: Substitute into equation (2).

$3x + 5\left(\frac{-2}{5}x + \frac{13}{5}\right) = 17$

Step 4: Expand, collect, and solve.

$3x + -2x + 13 = 17$

$x = 4 = 4$

Step 5: Substitute x back for y.

$y = \frac{-2}{5} \times 4 + \frac{13}{5} = 1 = 1$

Step 6: Check both original equations.

$2 \times 4 + 5 \times 1 = 13$ ✓

$3 \times 4 + 5 \times 1 = 17$ ✓

Teacher answers & working

Simultaneous Equations · Substitution Method

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E3

Solve the system:

$y = -5x - 3$

$6x + 4y = -54$

Answer:

x = 3,\; y = -18

Working:

Step 1: Write the system.

$$\begin{cases} y = -5x - 3 \\ 6x + 4y = -54 \end{cases}$$

Step 2: Substitute the expression for y into the second equation.

6x + 4\left(-5x - 3\right) = -54

Step 3: Expand and collect like terms.

6x + -20x - 12 = -54

-14x - 12 = -54

Step 4: Move the constant term and solve for x.

-14x = -54 + 12 = -42

x = 3 = 3

Step 5: Substitute x back into y = mx + b.

$y = -5 \times 3 - 3 = -18$

Step 6: Check in the second equation.

$6 \times 3 + 4 \times -18 = -54$ ✓